Osptek Display

AMOLED SPECIFICATION

Model No:

AM160Q320360ZS

Revision History

Version	Contents	Date	Note
00	Original	2025.6.10	

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1. Scope

This specification applies to the OLED module which is designed and manufactured by Shenzhen Osprey Optoelectronics Technology Co., Ltd.

2. Features

- 1.6” OLED
- Supported 320(RGB)*360
- Driver IC(1 Chip)：CO5300
- TP IC:CST820

3. General Specifications

No	Item	Specification	Unit
1	Display type	Rigid AMOLED	
2	Display Size	1.6”	Inch
3	Active Area	27.024 x 30.402 x 2.43	mm
4	Pixel Per inch	301	PPI
5	Resolution	320 X 360	
6	Color Depth	16.7M(24bit)	
7	Viewing Direction	All direction	
8	Pixel Aspect Ratio	1:1	
9	Frame rate	60HZ	
10	Driver IC (Type)	C05300	
11	Driver IC/ RAM Size	1/2RAM	
12	Panel bonding type	COF	
13	Interface	QSPI	
14	TFT Technology Type	LTPS	
15	TP IC	CST820	

4. Electrical Specifications

4.1 Absolute Maximum Ratings

Parameter	Symbol	Spec			Unit	Note
		Min.	Typ.	Max.		
Analog/boost power voltage	VCI	-0.3	-	5.5	V	-
I/O voltage	VDDIO	-0.3	-	5.5	V	-
Operating temperature	Top	-20	-	70	℃	-
Storage temperature	Tstg	-40	-	80	℃	-

4.2 DC Characteristics for Panel Driving

Power Characteristic:

Item	Symbol	Min.	Typ.	Max.	Unit	Remark
AMOLED Power Positive	ELVDD	3.25	3.3	3.35	V	-
AMOLED power Negative	ELVSS	-3.35	-3.3	-3.25	V	Ref
Digital Power supply	VDDIO	1.65	1.8	3.35	V	Ref
Analog Power supply	VCI	3.25	3.3	3.35	V	Ref

1) Normal Mode

Power Supply: VDDIO =1.8V VCI=3.3V ELVSS=-3.3V ELVDD=3.3V

Frame Frequency: F_{frame} =60HZ @ 25degC, Brightness 500 nits, Command Mode,

Display Condition	Symbol	Min.	Typ.	Max.	Unit	Remark
100% Pixel On 500nits	IELVDD /ELVSS	-	21.6	26.10	mA	Ref
	IVCI	-	5.8	6.5	mA	Ref
	IVDDIO	-	5.5	5.8	mA	Ref

2) Idle Mode

Power Supply: VDDIO=1.8V VCI=3.3V

Frame Frequency: F_{frame} =15HZ @ 25degC, Brightness 50 nits,

Display Condition	Symbol	Min.	Typ.	Max.	Unit	Remark
10% Pixel On 50 nits	IELVDD /ELVSS	-	-	-	mA	Supplied by Driver IC
	IVCI	-	3.4	4.0	mA	Ref
	IVDDIO	-	0.8	1	mA	Ref

3) Deep Standby Mode

Display Condition	Symbol	Min.	Typ.	Max.	Unit	Remark
Deep Standby	IVCI	-	-	3	uA	-
	IVDDIO	-	-	3	uA	-

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4.3 Electrical properties

4.3.1 Direct current characteristic

PARAMETER	SYMBOL	MIN	MAX	UNIT
Operating Supply Voltage	V_{IN}	2.9	5.5	V
Operating Free-Air Temperature	T_A	−40	+85	°C

Supply Voltage Range

Table 2 (Ta = -20°C~+60°C)

Item	Symbol	Unit	Min.	Typ.	Max.	Test condition
Power supply voltage	IOVCC	V	1.65	2.80	3.60	-
	AVCI	V	1.65	1.80	IOVCC or VCI	-
	VCI	V	2.5	2.80	3.60	-

*POWER SUPPLY * VBAT 2.9 V — 5.5 V

4.3.2 Photoelectric characteristic

V_{IN}=3.7V, AVDD=2.8V, OVDD=4.6V, OVSS=-2.4V, T_A=25°C, unless otherwise specified.

Parameter	Symbol	Test Condition	Min	Typ	Max	Unit
Input Power Supply						
Input Supply Voltage	V_{IN}		2.9	3.7	5.5	V
Quiescent Current	I_Q	SWIRE=High, AVDD_EN=High, measured into VIN pin. No load	-	70	-	µA
Standby Current	$I_{Standby}$	AVDD_EN =High and SWIRE = low	-	20		µA
Shutdown Current	I_{SHDN}	AVDD_EN and SWIRE = low	-	0.1	1	µA
Under-Voltage Lockout Threshold	V_{UVLOH}	VIN Rising	--	2.75	2.85	V
	V_{UVLOL}	VIN falling	--	2.5	2.6	V
Thermal Shutdown	T_{SD}		--	140	--	°C
Thermal Shutdown Hysteresis	ΔT_{SD}		--	10	--	°C
SWIRE						
SWIRE Logical High-Level Voltage	V_{SRH}	V_{IN}=2.9V to 5.5V	1.2	-	-	V
SWIRE Logic Low-Level Voltage	V_{SRL}	V_{IN}=2.9V to 5.5V	0	-	0.4	V
SWIRE Turn-off Detection	T_{OFF_DLY}		300	-	-	µs
SWIRE Signal Stop Indicate Time	T_{STOP}		300	-	-	µs
SWIRE Rising Time	T_r		-	-	200	ns
SWIRE Falling Time	T_f		-	-	200	ns
Clocked SWIRE High	T_{ON}		2	-	20	µs
Clocked SWIRE Low	T_{OFF}		2	-	20	µs
Input Clocked SWIRE Frequency	F_{SWIRE}		25	-	250	KHz
AVDDEN						
AVDD Enable Input Voltage	V_{IH}	V_{IN}=2.9V to 5.5V	1.2	-	-	V
	V_{IL}	V_{IN}=2.9V to 5.5V	0	-	0.4	V
SIBO						

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Positive Output Voltage Range	V_{OP}		2.7	-	5.4	V
Negative Output Voltage Range	V_{ON}		-4.8	-	-0.7	V
Switching Frequency	F_{SW}		1.2	1.5	1.8	MHz
Over Current Protection	I_{OCP}			0.75		A
AVDD LDO						
Positive Output Voltage Range	$V_{AVDD\ RANGE}$		2.6	2.8	3.5	V
Positive Output Voltage Accuracy	$V_{AVDD\ ACC}$		-1	-	1	%
Output Current Capability	I_{AVDD}		-	10	20	mA
Line Regulation	V_{AVDD_LINE}	V_{IN}=2.9 to 5.5V, I_{AVDD}=1mA	-	2	5	mV
Load Regulation	$V_{AVDD\ LOAD}$	I_{AVDD} = 0 to 10mA	-	2	5	mV
Output Ripple	$V_{AVDD\ RIPPLE}$	I_{AVDD} = 5mA	-	-	10	mV
Current Limit	$I_{AVDD\ LIMIT}$		-	30	50	mA
Discharge Resistance	$R_{AVDD\ RDIS}$			100		Ω
Under Voltage Protection				80		%
UVP Detection Time				1.35		ms
OVDD LDO						
Positive Output Voltage Range	$V_{OVDD\ RANGE}$		2.6	4.6	5.3	V
Positive Output Voltage Accuracy	$V_{OVDD\ ACC}$		-1	-	1	%
Output Current Capability	I_{OVDD}		-	80	100	mA
Line Regulation	$V_{OVDD\ LINE}$	V_{IN}=2.9 to 5.5V, I_{OVDD}=1mA	-	2	5	mV
Load Regulation	$V_{OVDD\ LOAD}$	I_{OVDD} = 0 to 10mA	-	2	5	mV
Output Ripple	$V_{OVDD\ RIPPLE}$	I_{OVDD} = 30mA	-	-	10	mV
Current Limit	$I_{OVDD\ LIMIT}$		-	150	200	mA
Discharge Resistance	$R_{OVDD\ RDIS}$		-	100	-	Ω
Under Voltage Protection				80		%
UVP Detection Time				1.35		ms
OVSS LDO						
Negative Output Voltage Range	$V_{OVSS\ RANGE}$		-4.7	-2.4	-0.6	V
Negative Output Voltage Accuracy	$V_{OVSS\ ACC}$		-1	-	1	%
Output Current Capability	I_{OVSS}		-	80	100	mA
Line Regulation	$V_{OVSS\ LINE}$	V_{IN}=2.7 to 5.5V, I_{OVSS}=1mA	-	2	5	mV
Load Regulation	V_{OVSS_LOAD}	I_{OVSS} = 0 to 10mA	-	2	5	mV
Output Ripple	$V_{OVSS\ RIPPLE}$	I_{OVSS} = 30mA	-	-	10	mV
Current Limit	$I_{OVSS\ LIMIT}$		-	150	200	mA
Discharge Resistance	$R_{OVSS\ RDIS}$			100		Ω
Under Voltage Protection				80		%
UVP Detection Time				1.35		ms

注：*Power IC 为BV6802,详细参考 "BV6802 Data Sheet

4.3.3 LCD photoelectric characteristics

Test condition：25℃±3℃，65±20%RH，darkroom，use the Auto-gamma system from EDO

No	Item		Symbol	Condition	Value			Unit	Remark
					Min.	Typ.	Max.		
1	Brightness		L	Full white Without CG	450	500	550	cd/m²	Note1
2	Brightness Uniformity		UL	Full white	85			%	Note4
3	Contrast Ratio		CR	Normal Θ=Φ=0°	10000	100000		-	Note3.
4	Response time		Ton+Toff	Normal Θ=Φ=0°		2	4	ms	Note2.
5	Color Coordinate of CIE1931	White	X	Normal Θ=Φ=0°	0.270	0.300	0.335	-	Note1.
			Y		0.280	0.310	0.345		
		Red	X		0.646	0.686	0.726		
			Y		0.27	0.310	0.35		
		Green	X		0.195	0.235	0.275		
			Y		0.68	0.720	0.76		
		Blue	X		0.103	0.143	0.183		
			Y		0.004	0.044	0.084		
6	Color Gamut		NTSC	CIE1931	97	108		%	
7	Viewing Angle			Top/Botton/Right/Left CR ratio ≥10	80			°	Note3.
8	Gamma			Log(Lv-Lb)=log(V)+log(a) V(Gray)= 48,72,104,132, 164,192,224 Lum(gray255)=500nit	1.9	2.2	2.5	-	
9	Flicker			Normal Θ=Φ=0°			-30	dB	Note6.
10	Crosstalk						6	%	Note7.
11	Color shift			θL=30°			6	JNCD	
12	OLED Life Time			0.95*(TYP brightness) At 25℃,with white color pattern	200			hrs	Note8.

10	Crosstalk						6	%	Note7.
11	Color shift			θL=30°			6	JNCD	
12	OLED Life Time			0.95*(TYP brightness) At 25℃,with white color pattern	200			hrs	Note8.

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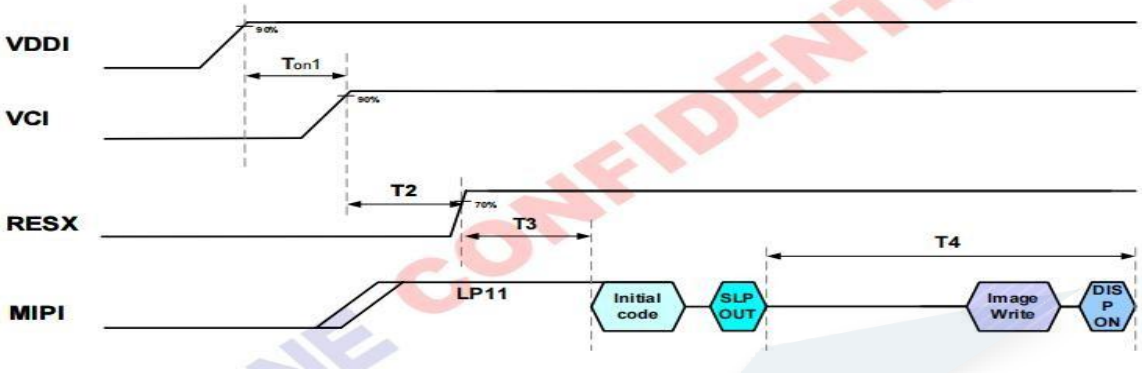
4.3.4 POWER ON/Of

The power on sequence for different power input modes are shown below figures.

Power ON Sequence Timing

Symbol	Description	Value			Unit	Remark
		Min.	Typ.	Max.		
Ton1	VDDI on to VCI on delay	>0			us	
T2	VDDI on to valid to RESET high	10			ms	
T3	RESET high to MIPI first commad	10			ms	
T4	Sleep-out command received to Display on command received.	120			ms	

The Power on sequence is shown as below.

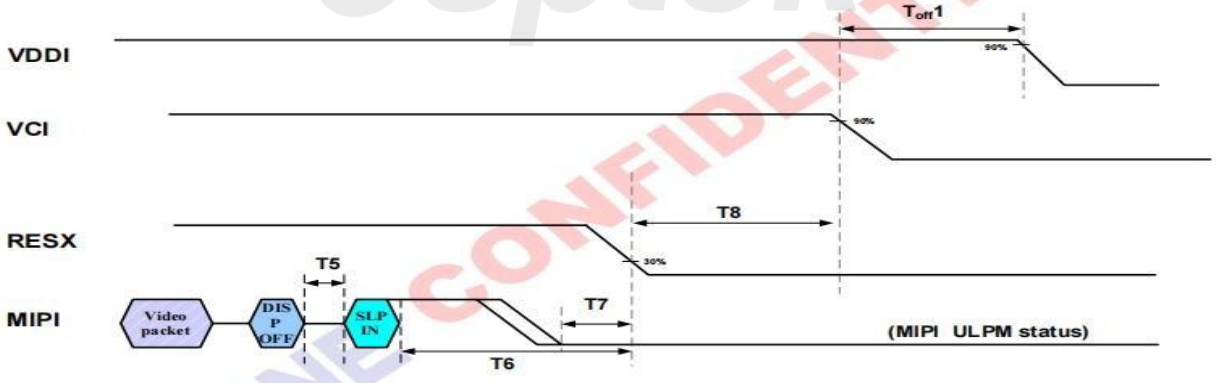


The power off sequence for different power input modes are shown below figures.

Power OFF Sequence Timing

Symbol	Description	Value			Unit	Remark
		Min.	Typ.	Max.		
Toff1	VCI off to VDDI off delay	>0			us	
T5	Display-off command received to Sleep-in command delay	>0			us	
T6	Sleep-in command received to valid to RESET low	100			ms	
T7	MIPI ultra low power mode to valid to RESET low	0			us	
T8	RESET low to VCI off delay	0			us	

The power off sequence is shown as below:



Power off sequence

Note 1: *Unless otherwise specified, timings herein show cross point at 50% of signal/power level.*

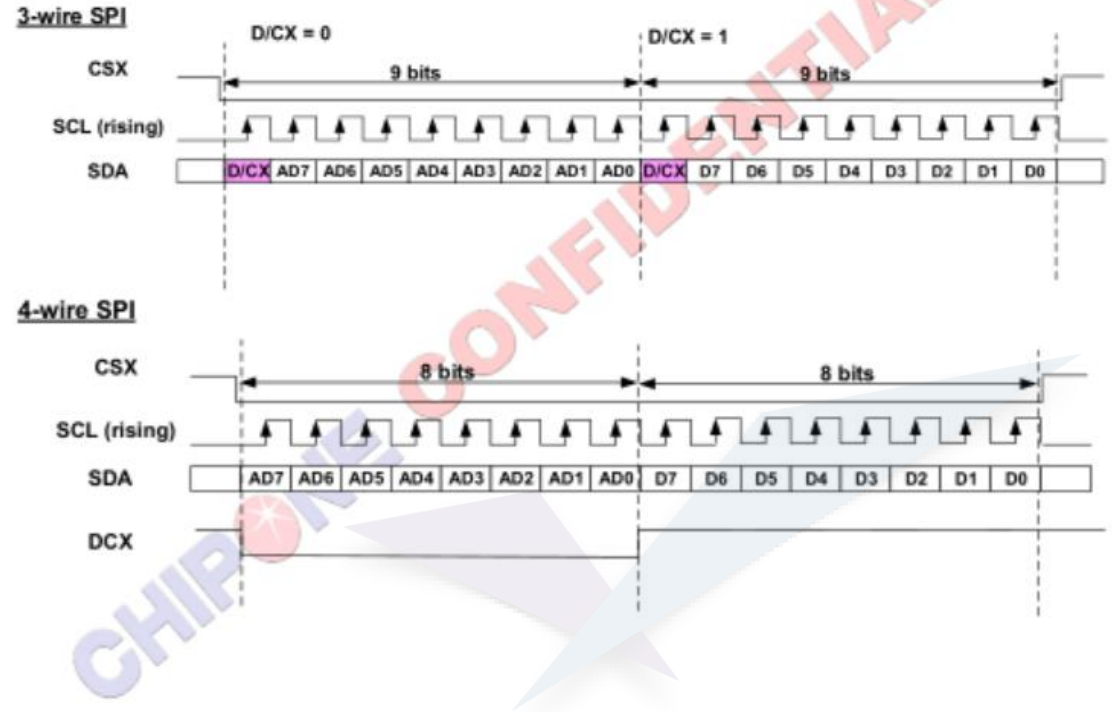
Note 2: *Keep VGH is equal to or larger than VCI during power off sequence.*

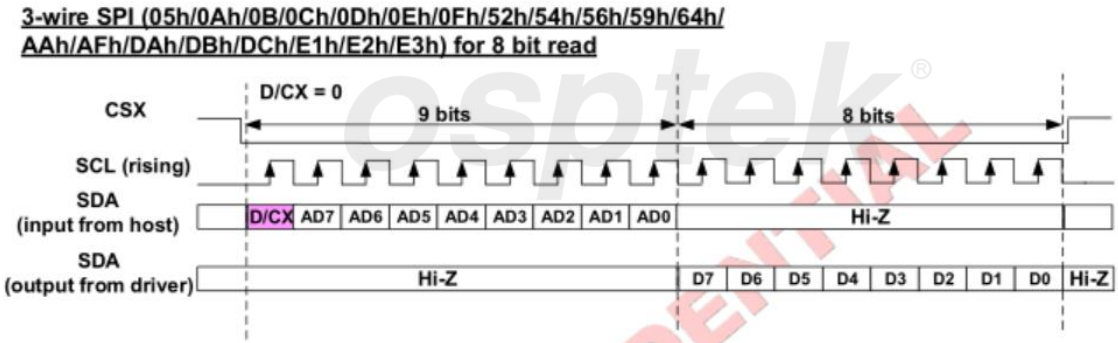
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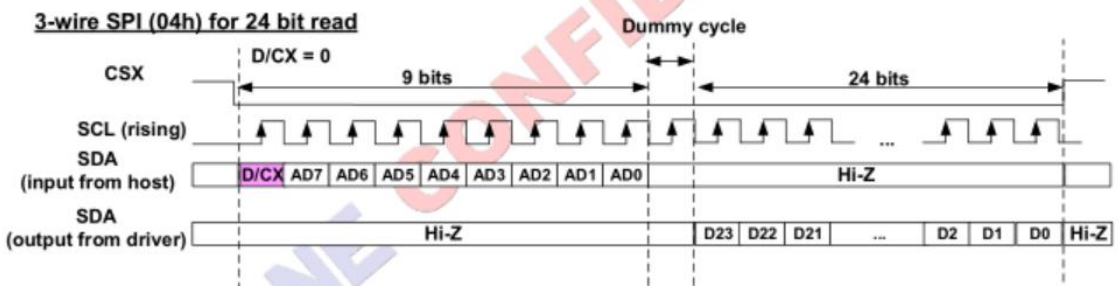
4.3.5 Write/Read Cycle and Sequence

The 3-wire/4-wire SPI interface utilizes DCX, CSX, SCL and SDA signals. SCL is driven from high to low then pulled back to high during the write cycle. The host processor provides information during the write cycle while the display module reads the host processor information on the rising edge of SCL.

The 3-wire/4-wire SPI interface write command sequences are described in the following figure as below.



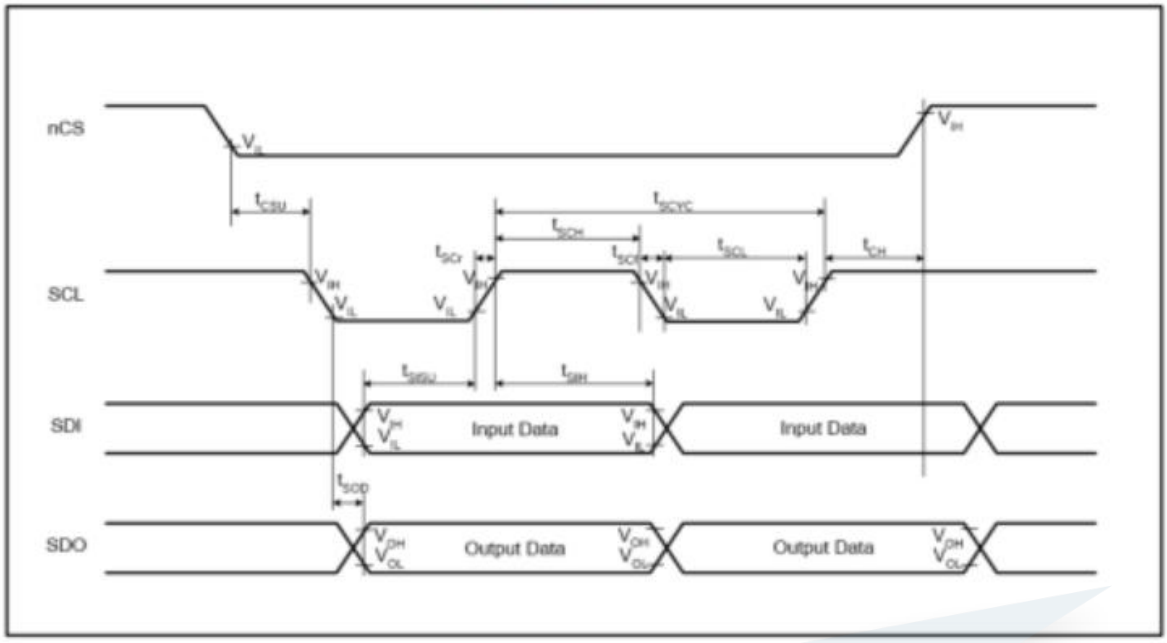




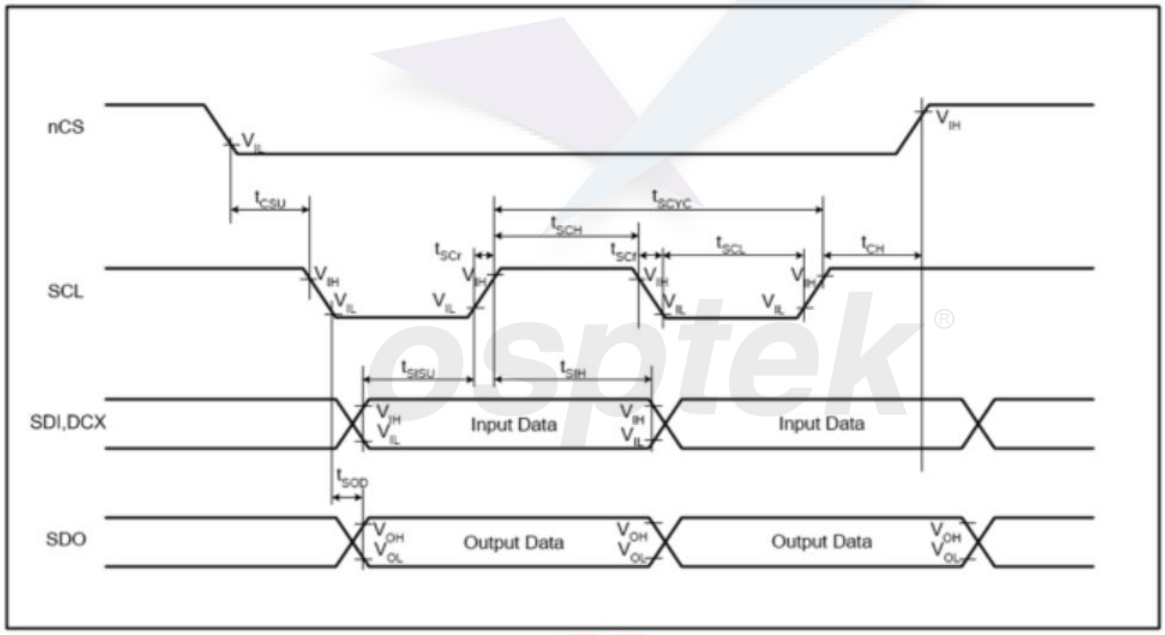
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4.4 AC Timings Characteristics

4-Wire SPI Serial Interface Characteristics



3-Wire SPI Serial Interface Characteristics



Parameter	Symbol	Conditions	Specification			Unit	Notes
			MIN	TYP	MAX		
SCL	T_{SCYC}	Clock cycle (Write)	20	-	-	ns	
	T_{SCYC}	Clock cycle (Read)	300		-	ns	
	T_{SCH}	Clock "H" pulse width (Write)	9	-	-	ns	
	T_{SCH}	Clock "H" pulse width (Read)	140	-	-	ns	
	T_{SCL}	Clock "L" pulse width (Write)	9	-	-	ns	
	T_{SCL}	Clock "L" pulse width (Read)	140	-	-	ns	
	T_{SCr}	Clock rise time	-	-	2	ns	
	T_{SCf}	Clock fall time	-	-	2	ns	
CSX	T_{CSU}	Chip select setup time	10	-	-	ns	
	T_{CH}	Chip select hold time	10	-	-	ns	
SDI (SDA)	T_{SISU}	Data input setup time	5	-	-	ns	
	T_{SIH}	Data input hold time	5	-	-	Ns	
SDO (SDA)	T_{SOD}	Data output setup time	-	-	120	ns	
	T_{SOH}	Data output hold time	5	-	-	ns	

***Note 1**: Logic high and low levels are specified as 20% and 80% of VDDI for Input signals.*

***Note 2**: Ta = -30 to 85 °C, VDDI=1.65V to 3.3V, VCI=2.7V to 3.6V, GND=0V*

5. Reliability Test Conditions And Methods

Item	Test Conditions	Remark
High Temperature Storage	Ta = 80℃ 120 hrs	
Low Temperature Storage	Ta =-30℃ 120 hrs	
High Temperature Operation	Ta = 70℃ 120 hrs	
Low Temperature Operation	Ta = -20℃ 120 hrs	
Storage at High Temperature and Humidity	+60℃, 90% 120 hrs	
Thermal Shock Test	-20℃ (30min) - +70℃(min); Cycles: 12times;	

6. Optical specifications

序号	项目		符号	条件	参数			单位	备注
					Min	Typ	Max		
1	Viewing angle	Left	ΘL	CR≥10	-	85	-	deg.	
		Right	ΘR		-	85	-		
		Top	ΦT		-	85	-		
		Bottom	ΦB		-	85	-		
2	Color Chromaticity (CIE1931)	White	Wx	θ=0°	0.27	0.30	0.33	-	
			Wy		0.28	0.31	0.34		
3	Luminance Uniformity		UL	θ=0°	85	-	-	%	
4	brightness		-	Center point	-	550	-	Cd/m2	

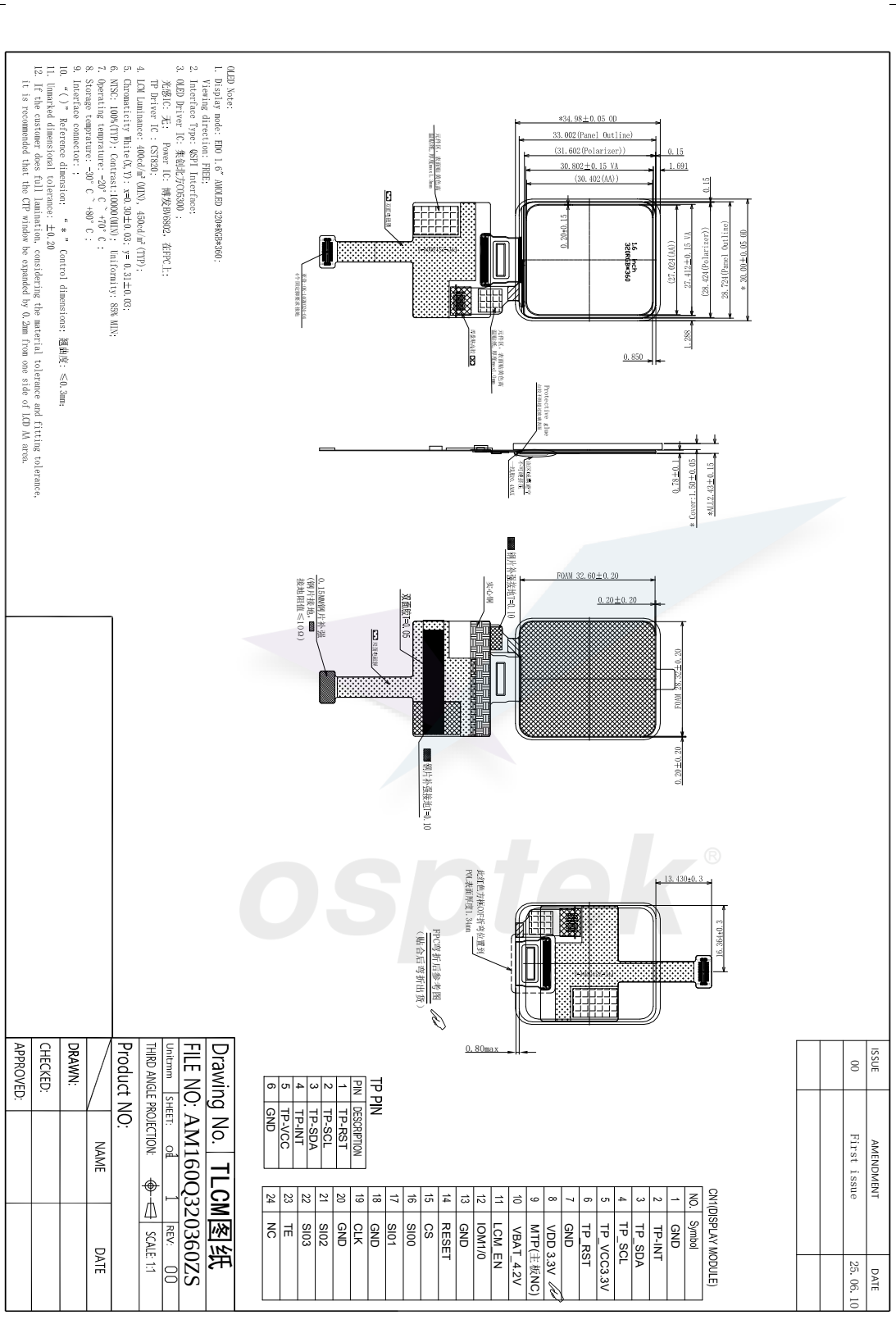
5	NTSC ratio	-		85	100		%	CIE1931
6	Cross talk	-	25℃			3	%	
7	Flicker	-	25℃			-30	DB	
8	Gamma	-		2.0	2.2	2.4		
9	Lifetime	T95	25℃	240			H	

7. Interface Pin Connection

Interfa ce NO.	Symbol	Description
1	GND	Ground
2	TP_INT	TP_INT
3	TP_SDA	TP_SDA
4	TP_SCL	TP_SCL
5	TP_VCC	TP_VCC 3.3V
6	TP_RST	TP_RST
7	GND	Ground
8	VDDI	Power supply for dispiay logic circuits 3.3V
9	MTP	MTP programming power supply pin(8.25TYP)/NC
10	VBAT	Power for power IC (3.7-5.5v)
11	EN	Power IC enable control pin, If not used, please open this pin.
12	NC	NC
13	GND	Ground
14	LCD REST	PCD Reset PIN
15	LCD_CS	Chip select input pin (“Low” enable) in 80-series MPU I/F and SPI I/F
16	LCD_D0	QSPI interface channel 0
17	LCD_D1	QSPI interface channel 1
18	GND	Ground
19	LCD_CLK	QSPI interface clock
20	GND	Ground
21	LCD_D2	QSPI interface channel
22	LCD_D3	QSPI interface channel
23	LCD_TE	Tearing effect output pin to synchronize MCU to frame writing, activated by S/W
24	NC	NC

8. Outline Dimension

TLCM 图纸



ISSUE	AMENDMENT	DATE
00	First issue	25.06.10

CN1(DISPLAY MODULE)

NO.	Symbol
1	GND
2	TP-INT
3	TP_SDA
4	TP_SCL
5	TP_VCC3.3V
6	TP_RST
7	GND
8	VDD 3.3V
9	MTP(主板NC)
10	VBAT_4.2V
11	LCM_EN
12	IOM1/0
13	GND
14	RESET
15	CS
16	SIO0
17	SIO1
18	GND
19	CLK
20	GND
21	SIO2
22	SIO3
23	TE
24	NC

TP PIN

PIN	DESCRIPTION
1	TP-RST
2	TP-SCL
3	TP-SDA
4	TP-INT
5	TP-VCC
6	GND

Drawing No. TLCM图纸

FILE NO: AM160Q320360ZS

Unit:mm	SHEET:	01	1	REV:	00
THIRD ANGLE PROJECTION:				SCALE: 1:1	

Product NO:

	NAME	DATE
DRAWN:		
CHECKED:		
APPROVED:		

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